Guidelines for Preparing Your CS Undergraduate Thesis for Publication

Publishing your undergraduate thesis in a reputable, Scopus-indexed academic journal or conference (Q3 or Q4 category) is a significant achievement. It transitions your work from a simple class requirement to a recognized contribution to the global scientific community.

To achieve this, your manuscript must shift from "telling me what you learned" to "showing the scientific community what you discovered." Below are the guidelines, focus areas, and structural templates to get your work published.

1. What to Focus on for Q3/Q4 Scopus Publication

Journals in the Q3 and Q4 quartiles are excellent targets for robust undergraduate research. They generally do not expect you to invent an entirely new branch of mathematics or a revolutionary new algorithm. Instead, they look for:

- **Sound Scientific Methodology:** Your experiments must be reproducible, well-documented, and logically sound.
- **Clear Application or Case Study:** Applying established techniques (like deep learning architectures or data analytics) to a new, localized problem or a newly gathered dataset is a highly successful strategy for Q3/Q4 papers.
- **Proper Evaluation:** Claims must be backed by rigorous testing. If you are training a model, you must show clear training/validation loss and accuracy curves, and use appropriate metrics (Precision, Recall, F1-Score) compared against baseline models.
- **Academic Tone and Formatting:** The paper must be written in formal, objective English, with properly formatted citations (usually IEEE or APA).

2. Mastering the Core Elements of Your Research

To pass peer review, reviewers will heavily scrutinize the following elements of your work:

The Research Problem & Objectives

- **The Problem:** Avoid vague statements like "Healthcare needs AI." Instead, define a specific, solvable technical gap. For example: "Current multi-class image classification systems struggle with identifying specific structural features and color variations in localized clinical nail datasets."
- **The Objectives:** Your objectives must be measurable. Use action verbs: *To evaluate, to compare, to optimize, to develop.*

Novelty and Contribution

Undergraduate students often panic about "novelty." For a Q3/Q4 publication, your novelty does not have to be a Nobel-prize-winning breakthrough. You can establish your contribution in a few realistic ways:

1. **Data Novelty:** Collecting and curating a new local dataset (e.g., gathering clinical data from local hospitals or regional clinics) and applying standard machine learning techniques to it.
2. **Application Novelty:** Applying existing models to a domain where they haven't been heavily utilized yet.
3. **Comparative Novelty:** Conducting a rigorous comparative analysis of multiple established backbones (e.g., VGG16, MobileNetV2, EfficientNetB0) on a specific complex problem to identify which performs best and *why*.

Methodology

This is the most critical section for peer review. If your method is flawed, the paper will be rejected regardless of how good the results look.

- Be explicit about your data preprocessing (handling missing values, data augmentation, normalization).
- Clearly define your system architecture. Use flowcharts and block diagrams.
- State your experimental setup: hardware used, hyperparameters, and validation techniques (e.g., k-fold cross-validation).

Results, Discussion, and Limitations

- **Results:** Present your findings objectively using tables and graphs. Do not just paste screenshots of code output; use professional plotting libraries (like Matplotlib) to create clean, high-resolution figures.
- **Discussion:** This is where you interpret the data. *Why* did EfficientNet outperform VGG16 in your specific case? How do your results compare to similar published studies?
- **Limitations: Do not skip this.** Acknowledging your limitations (e.g., a small dataset size, hardware constraints limiting epoch runs, or biases in data collection) shows academic maturity and actually protects your paper from harsh reviewer criticism.

3. Standard Manuscript Template

Most Scopus-indexed CS journals and conferences follow a variation of the IMRaD structure (Introduction, Methods, Results, and Discussion). Use the following outline for your manuscript draft:

Title: [Must be specific and descriptive. Avoid acronyms if possible.]

Abstract: [A strictly 150-250 word summary covering: Context, Problem, Proposed Solution/Method, Key Results, and Conclusion.]

Keywords: [5-7 specific terms used in your paper for search indexing.]

1. Introduction

- **Background:** Brief context of the broader field.
- **Problem Statement:** What specific gap or issue are you addressing?
- **Motivation:** Why is solving this problem important?
- **Contributions:** A bulleted list of 2-3 specific things this paper adds to the field. (e.g., "We introduce a newly curated dataset of...", "We evaluate four deep learning architectures on...")
- **Paper Organization:** A brief sentence outlining the rest of the paper.

2. Literature Review (or Related Work)

- Review 10-20 recent, relevant papers (preferably from the last 3-5 years).
- Do not just list papers ("Author A did X. Author B did Y."). Group them by theme and highlight the limitations in their work that *your* thesis is addressing.

3. Methodology

- **Dataset Description:** Source, size, features, and preprocessing steps.
- **Proposed Architecture/System Model:** Detailed explanation of the algorithms, neural network layers, or software architecture used.
- **Experimental Setup:** Software (e.g., Python, TensorFlow/PyTorch), hardware, and hyperparameters.
- **Evaluation Metrics:** The mathematical definitions of the metrics you are using to judge success.

4. Results

- **Performance Analysis:** Tables and charts displaying your core findings.
- **Comparative Analysis:** How does your proposed solution or best model stack up against baseline models or state-of-the-art literature?

5. Discussion

- Interpretation of the results.
- **Limitations:** Honest assessment of where the study falls short and boundaries of your system.

6. Conclusion and Future Work

- A concise summary of what was achieved and the final verdict of the research.
- **Future Work:** Logical next steps for researchers who might want to build upon your thesis.

References

- Strictly adhere to the target journal's citation style (usually IEEE or APA). Ensure every citation in the text matches the reference list, and prioritize citing peer-reviewed journal articles over websites.